

A Model Predictive Control Approach for Active Suspension based on a Half-car Model

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Abstract—A Model Predictive Control (MPC) approach is implemented for an active suspension system based on a half-car model. Two MPC controllers are designed and compared with a passive suspension system. Specifically, we develop a state-based MPC and an output-based MPC with reference tracking and disturbance rejection. Additionally, a stability analysis is conducted to verify the stability of the designed MPC controllers. Finally, numerical simulations are performed under various scenarios to evaluate the effectiveness of the proposed MPC controllers.

I. INTRODUCTION

The conventional vehicle suspension system, typically composed of a spring and a damper, connects the chassis to the road wheels to ensure both safety and comfort. Since the 1950s, active suspension systems have been a subject of ongoing research due to their potential to significantly enhance dynamic performance. However, their widespread adoption in production vehicles has long been hindered by high costs and technical complexity[1]. In recent years, advances in actuator technology and computing power have gradually mitigated these barriers, leading to an increasing number of vehicles being equipped with active suspension systems. This trend opens up opportunities for implementing more advanced functions, which in turn require the development of sophisticated control strategies.

Typical control strategies for active suspension include the Linear-Quadratic Regulator (LQR)[2] and Model Predictive Control (MPC)[3]. While LQR is computationally efficient, it often overlooks system constraints. In contrast, MPC optimizes control objectives while accounting for actuator limitations and ensuring stable wheel-road contact, ultimately enhancing ride comfort and vehicle performance.

In this project, a model predictive controller for an active suspension system is implemented on a typical C-Class vehicle[4], whose dynamics will be described in the following section.

In Section II, two types of MPC controllers are designed: a regulation MPC to drive the system to its equilibrium point and an output MPC with an observer, as the system state is not fully measurable and may be subject to disturbances. In Section III, the stability of the MPC controller is analyzed. Finally, several numerical simulations are conducted and discussed in Section IV.

A. Half-car Model Dynamics

The four-degree-of-freedom half-car model[5], as shown in Fig. 1, describes the vertical motion of a vehicle in response to

road disturbances. This model consists of three main components: the sprung mass, the front unsprung mass, and the rear unsprung mass. The sprung mass is connected to the unsprung masses through two suspension systems, which include active force actuators. The unsprung masses, in turn, are connected to the road through the tires, which are typically modeled as springs.

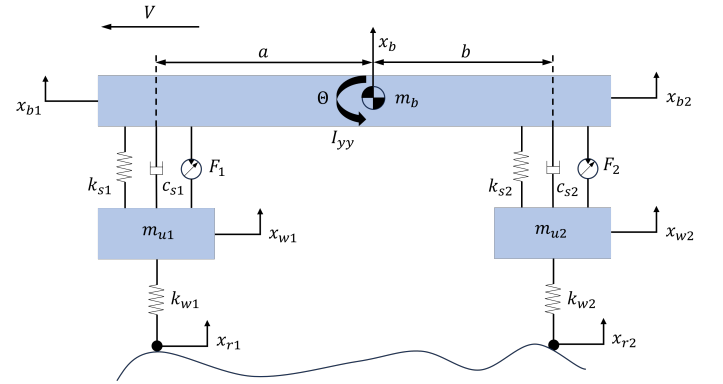


Fig. 1: Half-car Model

Applying Newton's Second Law, the equations of motion for the system can be expressed as follows:

$$\begin{cases} m_b \ddot{x}_b + k_{s1}(x_{b1} - x_{w1}) + c_{s1}(\dot{x}_{b1} - \dot{x}_{w1}) - F_1 \\ + k_{s2}(x_{b2} - x_{w2}) + c_{s2}(\dot{x}_{b2} - \dot{x}_{w2}) - F_2 = 0, \\ I_{yy} \ddot{\theta} - a[k_{s1}(x_{b1} - x_{w1}) + c_{s1}(\dot{x}_{b1} - \dot{x}_{w1}) - F_1] \\ + b[k_{s2}(x_{b2} - x_{w2}) + c_{s2}(\dot{x}_{b2} - \dot{x}_{w2}) - F_2] - M_y = 0, \\ m_{u1} \ddot{x}_{w1} + k_{w1}(x_{w1} - x_{r1}) - k_{s1}(x_{b1} - x_{w1}) \\ - c_{s1}(\dot{x}_{b1} - \dot{x}_{w1}) + F_1 = 0, \\ m_{u2} \ddot{x}_{w2} + k_{w2}(x_{w2} - x_{r2}) - k_{s2}(x_{b2} - x_{w2}) \\ - c_{s2}(\dot{x}_{b2} - \dot{x}_{w2}) + F_2 = 0, \\ x_{b1} = x_b - a\theta, \\ x_{b2} = x_b + a\theta. \end{cases} \quad (1)$$

where x_b , x_{w1} , and x_{w2} represent the vertical displacements of the sprung mass center of gravity, the front unsprung mass, and the rear unsprung mass, respectively. Similarly, x_{b1} and x_{b2} denote the vertical displacements at the front and rear axles. The pitch angle of the vehicle is given by θ .

The terms x_{r1} and x_{r2} correspond to the road excitations at the front and rear wheels. The masses of the sprung mass, front unsprung mass, and rear unsprung mass are denoted as m_b , m_{u1} , and m_{u2} , respectively.

The parameters k_{s1} , c_{s1} , k_{s2} , and c_{s2} represent the stiffness and damping coefficients of the front and rear suspension

systems, while k_{w1} and k_{w2} denote the vertical stiffness of the tires. The pitch inertia of the vehicle is given by I_{yy} .

The actuator forces at the front and rear axles are represented by F_1 and F_2 , respectively. M_y represents an external pitch moment generated by the longitudinal forces exerted by the tires. Finally, a and b denote the distances from the center of gravity to the front and rear axles.

TABLE I: Parameters of the Half-car Model

Symbol	Variable Name	Value	Unit
m_b	Sprung mass	690	kg
m_{u1}	Front unsprung mass	40.5	kg
m_{u2}	Rear unsprung mass 2	45.4	kg
k_{s1}	Front spring stiffness	17000	N/m
k_{s2}	Rear spring stiffness	22000	N/m
c_{s1}	Front damper damping	1500	Ns/m
c_{s2}	Rear damper damping	1500	Ns/m
k_{w1}	Front tire stiffness	192000	N/m
k_{w2}	Rear tire stiffness	192000	N/m
I_{yy}	Pitch moment of inertia	1222	kg · m ²
a	Distance from CoG to front axle	1.25	m
b	Distance from CoG to rear axle	1.51	m
F_{sat}	Maximum actuator force	3500	N

The parameters of the model are listed in Table I, which corresponds to a typical C-Class vehicle[4]. Based on Equation (1), a linear time-invariant (LTI) system can be derived as follows:

$$\begin{cases} \dot{X} = AX + BU \\ Y = CX \end{cases} \quad (2)$$

The C matrix represents the sensor configuration and can either be an identity matrix for state-feedback control or a subset of the system states, which will be specified in the output MPC section. The active suspension system consists of 8 states and 5 control inputs. The states and control inputs are:

- $\dot{x}_b$: Vertical velocity of sprung mass
- $\dot{\theta}$: Pitch rate of sprung mass
- $\dot{x}_{w1}, \dot{x}_{w2}$: Vertical velocity of front and rear unsprung mass
- $x_{b1} - x_{w1}$: Front suspension travel
- $x_{b2} - x_{w2}$: Rear suspension travel
- $x_{w1} - x_{r1}$: Front tire deflection
- $x_{w2} - x_{r2}$: Rear tire deflection
- F_1, F_2 : Front and rear suspension active force
- $\dot{x}_{r1}, \dot{x}_{r2}$: Vertical velocity of road excitations
- M_y : External pitch moment

$$X = \begin{bmatrix} \dot{x}_b \\ \dot{\theta} \\ \dot{x}_{w1} \\ \dot{x}_{w2} \\ x_{b1} - x_{w1} \\ x_{b2} - x_{w2} \\ x_{w1} - x_{r1} \\ x_{w2} - x_{r2} \end{bmatrix} \quad U = \begin{bmatrix} F_1 \\ F_2 \\ \dot{x}_{r1} \\ \dot{x}_{r2} \\ M_y \end{bmatrix} \quad (3)$$

The continuous-time dynamics are discretized using a sampling time of 0.01s, ensuring that the resulting discrete-time system remains both controllable and observable. The

discretization is performed using the zero-order hold (ZOH) method, which accurately simulates the behavior of typical digital-to-analog (DAC) and analog-to-digital (ADC) converters commonly used in such applications.

B. Road Bump Model

The primary source of vehicle vertical motion excitation is the road x_{r1} and x_{r2} . Therefore, a bump road model is developed in this section to simulate the vehicle's response after driving over a bump and to evaluate how the controller helps regulate the vehicle's vertical motion. The shape of the bump is described as follows.

$$\begin{cases} x_{r1} = \frac{A}{2} \left(1 - \cos \frac{2\pi V(t-t_0)}{L} \right), & t_0 \leq t \leq t_0 + \frac{L}{V} \\ x_{r2} = \frac{A}{2} \left(1 - \cos \frac{2\pi V(t-t_1)}{L} \right), & t_1 \leq t \leq t_1 + \frac{L}{V} \\ t_1 = t_0 + \frac{a+b}{V} \end{cases} \quad (4)$$

where A denotes the bump height, L represents the bump length, and V is the vehicle's longitudinal speed. The parameters t_1 and t_2 denote the times when the front and rear tires encounter the bump, respectively, while t represents the simulation time. Also, the vertical speeds, which are inputs to the system, can be obtained by taking the derivative of the equations.

II. MODEL PREDICTIVE CONTROL DESIGN

Based on the dynamics described in I, two different MPC controllers were designed to control the active suspension system:

- 1) Regulation MPC using full-state feedback
- 2) Output MPC using a subset of the states with disturbance rejection

A. Regulation MPC

In this case, full-state information is assumed and the goal is to stabilize the discretized system as described in Equation 2. The equilibrium at origin represents that the car is moving at a constant speed on an absolutely level road with no body vibration, pitch angle, and tire deflection. The cost function used to solve the MPC problem at each iteration is defined as follows:

$$J(x_0, \mathbf{u}_N) = \sum_{k=0}^{N-1} \ell(x(k), u(k)) + V_f(x(N)) \quad (5)$$

$s.t. \mathbf{u}_N \in \mathbb{U}, x \in \mathbb{X}, x(N) \in \mathbb{X}_f$

with the stage and terminal costs defined as:

$$\begin{aligned} \ell(x(k), u(k)) &= x(k)^T Q x(k) + u(k)^T R u(k) \\ V_f(x(N)) &= x(N)^T P x(N) \end{aligned} \quad (6)$$

where $Q = \text{diag}(10^4, 10^4, 10^4, 10^4, 10^5, 10^5, 10^4, 10^4)$, $R = 10^{-4}I$ and P is the solution of the discrete algebraic Riccati equation (DARE) used to solve the unconstrained infinite-horizon LQR problem. The choices of Q and R were obtained following an iterative tuning procedure which could

ensure a balance between control action and settling time, which is detailed illustrated in section IV. P is the DARE solution and X_f was chosen properly as it would help the stability proof in III. Note that Q , R and P are positive definite.

The inputs consist of two actuator forces, two road excitations, and one external pitch moment. Since the actuator has an output saturation, the force generated by the actuator is bounded by $3500N$. Additionally, to demonstrate the system's asymptotic stability in section III, it is assumed that the road inputs and pitch moment are zero at this point. This assumption is reasonable because, if these inputs persist over time, the system would continuously be driven out of its equilibrium point. The input constraints $\mathbf{u}_N \in \mathbb{U}$ in Equation 5 are therefore defined as:

$$\begin{aligned} |F_1| &= |\mathbf{u}_N(1:5:end)| \leq F_{sat} = 3500 \\ |F_2| &= |\mathbf{u}_N(2:5:end)| \leq F_{sat} = 3500 \\ \dot{x}_{r1} &= \mathbf{u}_N(3:5:end) = 0 \\ \dot{x}_{r2} &= \mathbf{u}_N(4:5:end) = 0 \\ M_y &= \mathbf{u}_N(5:5:end) = 0 \end{aligned} \quad (7)$$

Meanwhile, two areas of state constraints should be considered[3] to ensure that the active suspension system operates within its intended scope:

- 1) Suspension deflection: Since the suspension system is positioned between the vehicle body and the wheels, structural constraints exist on the mechanical components. Therefore, the suspension deflection limit must be taken into account during controller design to prevent bottom-out damage.
- 2) Road-holding limit: To ensure driving safety, the wheel must maintain continuous firm contact with the road at all times.

$$\begin{aligned} |x_{b1} - x_{w1}| &\leq SS_{max} \\ |x_{b2} - x_{w2}| &\leq SS_{max} \\ k_{w1}(x_{w1} - x_{r1}) &\leq \left(\frac{m_b b}{a+b} + m_{u1}\right)g \\ k_{w2}(x_{w2} - x_{r2}) &\leq \left(\frac{m_b a}{a+b} + m_{u2}\right)g \end{aligned} \quad (8)$$

where SS_{max} is limiting constraint amount that the suspension movement must not exceed, and g is the acceleration of gravity. These constraints can be represented as follows:

$$|x_i(k)| \leq x_{i,limit} \quad \forall i \in \{5, 6, 7, 8\}, k \in [0, \dots, N] \quad (9)$$

Since the MPC problem is an optimization on the control inputs, the state constraints are related to the input constraints by using the prediction relationship $\mathbf{x}_N = T\mathbf{x}_0 + S\mathbf{u}_N$. The state constraints can be rewrite as:

$$\begin{aligned} |T\mathbf{x}_0 + S\mathbf{u}_N| &\leq x_{limit} \\ \Rightarrow \begin{cases} S\mathbf{u}_N \leq -T\mathbf{x}_0 + x_{limit} \\ S\mathbf{u}_N \geq -T\mathbf{x}_0 - x_{limit} \end{cases} \end{aligned} \quad (10)$$

In case of the prediction horizon, N was set to 8 to ensure system controllability while keeping the MPC problem

computationally manageable. The detailed tuning process will be presented in Section IV.

B. Output MPC and Disturbance Rejection

In real-world applications, full-state measurements are often impractical. To simulate output-based control, an output MPC controller was designed. The sensor array consists of an onboard IMU mounted on the sprung mass and a linear displacement sensor inside the actuator. This setup allows the following measurements:

- 1) Sprung mass pitch rate $\dot{\theta}$
- 2) Linear displacement of the suspension $x_{b1} - x_{w1}$, $x_{b2} - x_{w2}$

To incorporate disturbance rejection into the closed-loop system, an augmented system is introduced to estimate and compensate for disturbances affecting the system. The augmented state-space representation is given by:

$$\begin{aligned} \begin{bmatrix} \dot{x}^+ \\ \dot{d}^+ \end{bmatrix} &= \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} \begin{bmatrix} x \\ d \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u + \begin{bmatrix} w \\ w_d \end{bmatrix} \\ y &= \begin{bmatrix} C & C_d \end{bmatrix} \begin{bmatrix} x \\ d \end{bmatrix} + v \end{aligned} \quad (11)$$

where the matrices B_d and C_d are used to represent the affect of a constant disturbance on the x and y states. In this case, we simulate a constant aerodynamic lift force acting on the vehicle at high speed, as well as a measurement error in the suspension's linear displacement due to fluctuations in the reference voltage. To ensure that the augmented system is observable, it is required that the original (A, C) system is observable as well as:

$$\text{rank} \begin{bmatrix} I - A & -B_d \\ C & C_d \end{bmatrix} = n + n_d \quad (12)$$

In this case, this equation is satisfied with $n + n_d = 10$. Additionally, a Luenberger observer is designed to ensure rapid convergence of the estimated disturbance to its actual value. For reference tracking and disturbance rejection, an optimal target selection (OTS) needs to be solved online to obtain the required x_{ref} and u_{ref} to reject disturbances on the states. The OTS problem to be solved online is described as:

$$(x_{ref}, u_{ref})(\hat{d}, y_{ref}) \in \begin{cases} \arg \min_{x_r, u_r} J(x_r, u_r) \\ s.t. \\ \begin{bmatrix} I - A & -B \\ C & 0 \end{bmatrix} \begin{bmatrix} x_r \\ u_r \end{bmatrix} = \begin{bmatrix} 0 \\ y_{ref} - \hat{d} \end{bmatrix} \\ (x_r, u_r) \in \mathbb{Z} \\ Cx_r + \hat{d} \in \mathbb{Y} \end{cases} \quad (13)$$

With the references x_{ref} and u_{ref} solved using the OTS at each iteration, the stage cost and terminal cost are adjusted to account for the reference signals as follow:

$$\begin{aligned} \ell(x(k), u(k)) &= (x(k) - x_{ref})^T Q (x(k) - x_{ref}) \\ &\quad + (u(k) - u_{ref})^T R (u(k) - u_{ref}) \\ V_f(x(N)) &= (x(N) - x_{ref})^T P (x(N) - x_{ref}) \end{aligned} \quad (14)$$

Following a similar procedure as the regulation MPC, the cost weighing matrices were chosen to be $Q = I$, $R = 10^{-2}I$, and P is the solution to the DARE of the unconstrained problem. Finally, the prediction horizon was chosen to be 8 for the same reason as described in the regulation MPC part.

III. ASYMPTOTIC STABILITY

In this section, we show that the designed MPC asymptotically stabilized the closed-loop system provided the system's initial conditions are sufficiently close to the origin.

We will follow the approach of Section 2.5.4 of [6], which discusses the linear MPC case with quadratic cost functions and constraints, to show that the aforementioned system design is exponentially stable in $\mathcal{X}_n$ based on Assumptions 2.2, 2.3 and 2.14 in [6], by Theorems 2.19 and 2.21. The stability of the Output MPC will be verified in section IV since it is hard to check a-priori. The following sub-sections are the procedures to prove system stability.

A. Assumption 2.2

The system described in Equation 2 is linear, which means that it is continuous and has an equilibrium point as $(x, u) = (0, 0)$. Also, the stage cost and terminal cost defined in Equation 5 are continuous and positive definite with $\ell(0, 0) = 0$ and $V_f(0) = 0$.

B. Assumption 2.3

The state constraints and input constraints described in Equation 10 and Equation 7 are both closed and contain the origin. The input constraints are compact. Also, the terminal set $\mathbb{X}_f \in \mathbb{X}$ is compact and contains the origin, which will be shown in the following section.

C. Assumption 2.14

Assumption 2.14 (a) states that the terminal cost is a Control Lyapunov Function (CLF), which is verified if $\forall \in \mathbb{X}_f, \exists u \in \mathbb{U} s.t. f(x, u) \in \mathbb{X}_f$ and $V_f(f(x, u)) - V_f(x) \leq -\ell(x, u)$.

In the regulation MPC, the weighing matrix P is obtained from the DARE solution $A_k^T P A_k = P - Q_k$. Here, $A_k = A + BK$ with K the solution to the DARE, and $Q_k = Q + K^T R K$. With $u = Kx$, we can get the following equation:

$$\begin{aligned} V_f(A_k x) &= x^T P x - x^T Q_k x \\ &= x^T P x - x^T Q_k x \\ &= V_f(x) - \ell(x, u) \end{aligned} \quad (15)$$

The Lyapunov decrease is thus proven for an initial condition with $\mathbb{X}_f$.

Assumption 2.14 (b) can be verified by $\ell(x, u) \geq q\|x\|^2 + r\|u\|^2$ and $V_f(x) \leq p\|x\|^2$ for $q = 10^4$, $r = 10^{-4}$ and $p = \lambda_{max}(P) = 2.161 \times 10^6$.

D. Estimating $\mathbb{X}_f$ and $\mathcal{X}_n$

We have proven the stability of the system above. The goal of this section is to find the maximum invariant constraint-admissible set $\mathbb{X}_f$ and the region of attraction $\mathcal{X}_N$, such that initial conditions starting in $\mathcal{X}_N$ are guaranteed to converge asymptotically to the origin.

The set $\mathbb{X}_f$ ensures that the MPC optimal control input coincides with the unconstrained control input from the infinite-horizon LQR problem. Within this set, the optimal MPC feedback law satisfies $\mathcal{K}_N(x) = Kx$, where K is the optimal LQR gain.

Using the algorithm presented in Exercise 41, an invariant set $\mathbb{X}_f$ can be estimated, ensuring compliance with both control input and state constraints. This algorithm computes a state and input admissible invariant set containing all states where the unconstrained LQR control law keeps the system trajectories within the set. The details of this algorithm are provided in the Appendix. Using this approach, the set $\mathbb{X}_f$ is found to contain 60 linear inequalities represented by $Gx \leq g$. $\mathbb{X}_f$ is formally defined as:

$$\mathbb{X}_f = \{x \in \mathbb{X}^8 \mid Gx \leq g\} \quad (16)$$

With an estimation of $\mathbb{X}_f$, an empirical analysis is conducted to approximate the region of attraction $\mathcal{X}_N$. This is achieved by considering a large set of initial conditions x_0 and verifying the feasibility of the MPC problem with the terminal constraint $x(N) \in \mathbb{X}_f$. If the problem is feasible for a given x_0 , then $x_0 \in \mathcal{X}_N$.

Since the system has 8 states, $\mathcal{X}_N$ cannot be visualized directly. However, a 2-D projection of $\mathcal{X}_N$ can be represented, and similar projections can be obtained for other state combinations. This approach allows us to validate whether a specific initial condition belongs to $\mathcal{X}_N$.

As can be seen in Fig. 2, a larger prediction horizon N will result in a larger region of attraction $\mathcal{X}_N$.

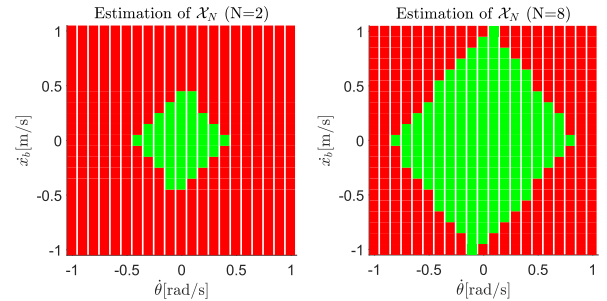


Fig. 2: 2-D projection of the estimated $\mathcal{X}_N$ for different initial $\dot{x}_b$ and $\dot{\theta}$

IV. NUMERICAL SIMULATIONS

In this section, we run several numerical simulations where we compare some MPC controllers as well as typical functions of the active suspension.

A. Prediction Horizon N

The system response for different prediction horizons is illustrated in Fig. 3. Notably, when $N = 2$, the system becomes unstable. A larger prediction horizon results in a reduced settling time, with $N = 5$ demonstrating a stable response. Moreover, increasing N expands the region of attraction $\mathcal{X}_N$, as discussed in the previous section. Considering this, the prediction horizon was set to $N = 8$ to ensure both stability and a sufficiently large region of attraction.

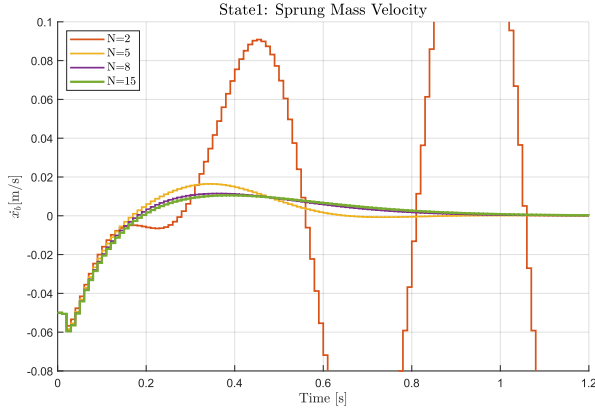


Fig. 3: System response for different prediction horizons N

B. Regulation MPC with Different Cost Function Weightings

In this subsection, regulation MPC is employed to demonstrate how varying cost function weightings influence the performance of the closed-loop system.

Figure 4 illustrates the stabilization of the suspension system under different values of Q . To facilitate a clearer comparison of the impact of various weightings, four key states are selected: sprung mass velocity $\dot{x}_b$, sprung mass pitch rate $\dot{\theta}$, front suspension travel $x_{sus1} = x_{b1} - x_{w1}$, and rear suspension travel $x_{sus2} = x_{b2} - x_{w2}$. These states are chosen as they are of greater concern compared to tire deflections and unsprung mass velocity.

As observed in Figure 4, smaller state cost values lead to longer settling times and greater overshoot in suspension travel. However, when Q is excessively large, the states may oscillate slightly around the equilibrium point. This phenomenon can be attributed to the controller's heightened sensitivity to state errors, requiring larger actuator forces to drive the system toward equilibrium. Consequently, an appropriate value of Q is carefully chosen to strike a balance between these effects.

Figure 5 presents the state trajectories for different values of R while maintaining the base state cost Q . As observed in the figure, a smaller R leads to a shorter settling time and reduced overshoot. However, slight oscillations may occur when R is too small.

Additionally, Figure 6 illustrates that with smaller R values, the controller applies larger forces to the system. This behavior aligns with the state trajectories and can be explained by the fact that the controller imposes less penalty on large control efforts.

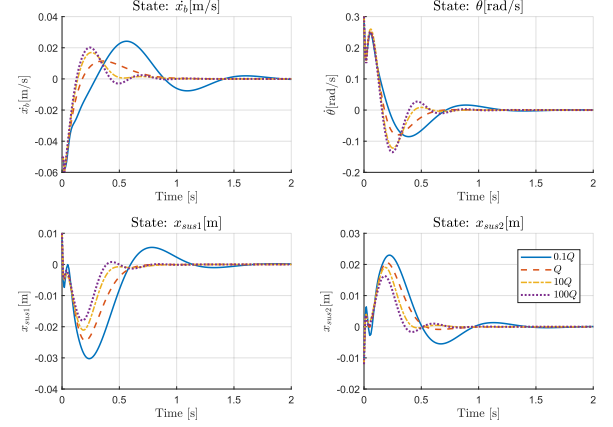


Fig. 4: State trajectories for different stage cost matrices Q with $R = R_{base}$ and $N = 8$

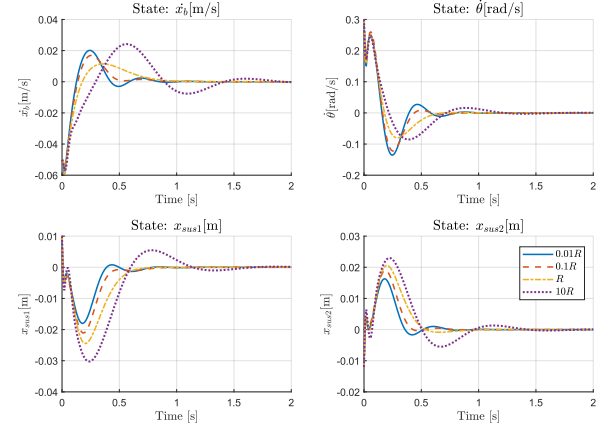


Fig. 5: State trajectories for different stage cost matrices R with $Q = Q_{base}$ and $N = 8$

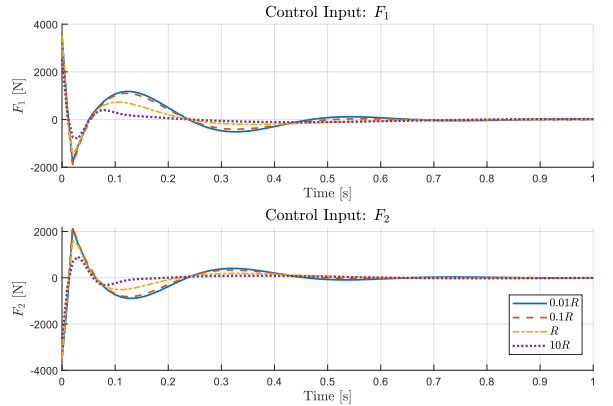


Fig. 6: Input actuator forces trajectories for different stage cost matrices R with $Q = Q_{base}$ and $N = 8$

C. Regulation MPC with Bump Road and Pitch Moment

In this subsection, a scenario is set up to evaluate the performance of regulation MPC and compare it with a passive suspension system, i.e., a suspension without actuators.

Initially, the vehicle starts from an initial state of

$$x_0 = [-0.05; 0.3; 2; -2; 0.01; -0.012; 0.002; -0.002]$$

with a constant velocity of 30 km/h. At $t = 5$ s, the front and rear axles successively encounter a bump, whose shape is defined by Equation 4 with parameters $L = 10$ m and $A = 0.05$ m.

Then, at $t = 10$ s, an external pitch moment is applied to the sprung mass due to the longitudinal force generated by the vehicle as it decelerates to a standstill. This external pitch moment is computed as

$$M_y = F_x h_{CoG} = 1940 \text{ Nm}$$

which corresponds to a deceleration of approximately $0.3g$.

As shown in Figure 7, the regulation MPC stabilizes the sprung mass within 2 seconds after encountering a bump, guiding the system smoothly to the equilibrium point compared to the passive suspension system.

Furthermore, after applying an external pitch moment, the regulation MPC generates control forces on the sprung mass, as illustrated in Figure 8. This helps to minimize suspension travel, leading to a smaller pitch angle. As a result, passengers experience greater comfort during vehicle acceleration and braking.

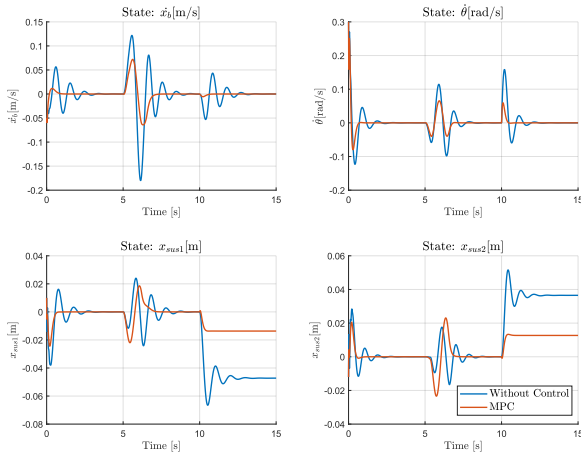


Fig. 7: State trajectories for regulation MPC compared to a passive suspension (without control)

D. Output MPC and Disturbance Rejection using Observer

In this subsection, we assume that only the sprung mass pitch rate $\dot{\theta}$, front suspension travel x_{sus1} , and rear suspension travel x_{sus2} are available for measurement. Additionally, two types of disturbances affect the suspension system:

- 1) d_1 : An external aerodynamic lift force acting on the sprung mass.

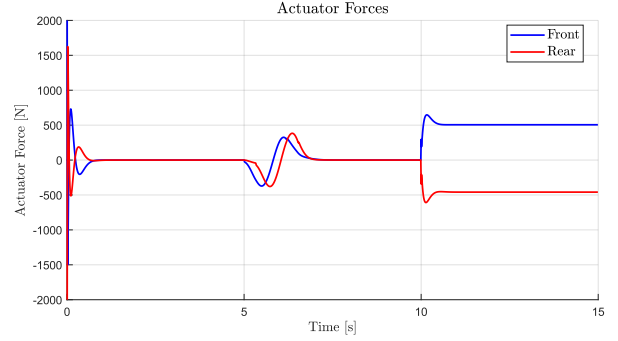


Fig. 8: Input actuator forces trajectories for regulation MPC

- 2) d_2 : A measurement error in the rear suspension travel due to overvoltage in the reference voltage.

As described in Section II, an observer is implemented to estimate these disturbances.

Figure 9 illustrates a simulation in which disturbance d_1 affects the suspension system during two distinct time intervals: $t = [1s, 2s]$ and $t = [8s, 15s]$. Meanwhile, another disturbance, d_2 , is applied to the system during $t = [10s, 15s]$. Additionally, both the front and rear suspensions are required to lift the sprung mass by approximately 50 mm starting from $t = 5s$.

In real-world scenarios, when a potential crash is detected, active suspension systems often raise the vehicle body to align the bumper beam with the impact, allowing it to absorb more energy and enhance passenger safety.

From the figure, it is evident that the output MPC controller effectively rejects disturbances. Furthermore, when tracking a step reference, the system achieves a rise time of approximately 100 ms, demonstrating exceptionally fast response characteristics.

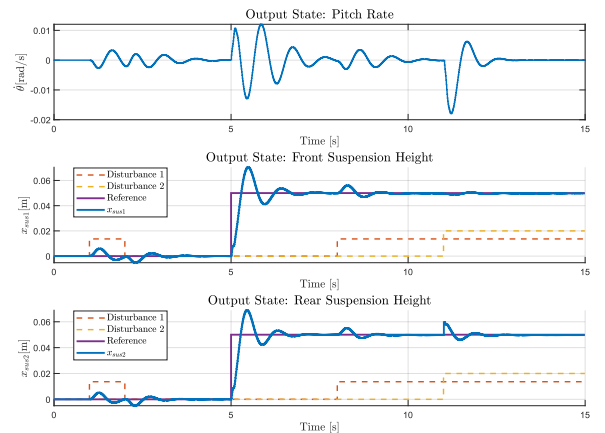


Fig. 9: Reference tracking and disturbance rejection using Output MPC

Figure 10 presents the estimated disturbances from the observer acting on the suspension system. The observer accurately detects the disturbances within approximately 200 ms, which is significantly faster than the system's settling time.

Furthermore, as discussed in Section III, since the observer reliably identifies disturbances, the output MPC remains stable, which is verified *a posteriori*.

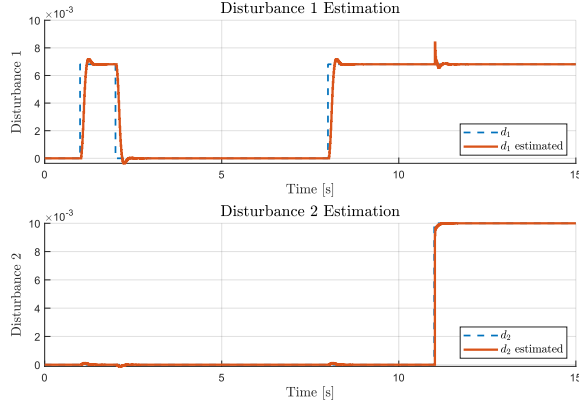


Fig. 10: Estimated disturbances acting on the suspension system as well as the actual step disturbance

Figure 11 shows the changes in reference states x_r and reference inputs u_r determined by the OTS problem during the whole process.

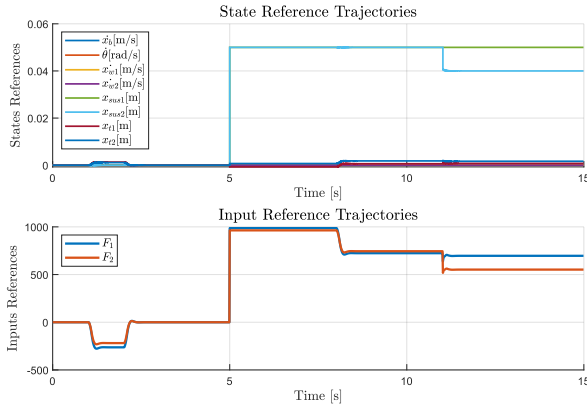


Fig. 11: Reference states x_r and inputs x_r determined by the OTS optimization under influence of disturbances

V. CONCLUSION

In this paper, a Model Predictive Control (MPC) approach was implemented and evaluated for an active suspension system based on a half-car model. Two MPC controllers—state-based MPC and output-based MPC—were designed and compared with a passive suspension system. The proposed controllers incorporated reference tracking and disturbance rejection to enhance ride comfort and vehicle stability. Also, the stability of the proposed controller is verified.

Simulation results demonstrated that both MPC controllers significantly outperformed the passive suspension system in reducing vibration and improving ride quality. Notably, the output-based MPC exhibited superior adaptability by actively adjusting the ride height in 100ms, which contributes to enhanced collision protection and improved vehicle dynamics in varying road conditions.

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APPENDIX

Algorithm 1: Estimate Set $\mathbb{X}_f$

Result: $Hx \leq h$ representing $\mathbb{X}_f$

Initialization:

$A_K := A - BK$

$K := \text{LQR optimal gain}$

$K_{\text{aug}} := [K; I]$

Set $k := 0$

for $i = 1, 2, \dots, s$ **do**

$x_i^* := \arg \max_x f_i(K_{\text{aug}} A_K^{k+1} x)$
s.t. $f_j(K_{\text{aug}} A_K^t x) \leq 0, \forall j \in \{1, 2, \dots, s\},$
 $\forall t \in \{0, 1, \dots, k\}$

end

if $f_i(K_{\text{aug}} A_K^{k+1} x) \leq 0, \forall i \in \{1, 2, \dots, s\}$ **then**

$\mathbb{X}_f := \{x \in \mathbb{R}^n \mid f_i(K_{\text{aug}} A_K^t x) \leq 0, \forall j \in \{1, 2, \dots, s\}, \forall t \in \{0, 1, \dots, k\}\};$

end

else

Set $k := k + 1$ and continue;

end